

ELECTRICAL & ELECTRONICS ENGINEERING

Manipal Institute of Technology

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ACTIVE SUSPENSION SYSTEM Design and Performance Analysis

Systems Simulation Laboratory

ELE 3142

5th Semester B.Tech. (E & E Engineering)

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PROJECT SYNOPSIS

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Abstract

This project presents the design, analysis, and simulation of an active suspension system for automotive applications using a quarter-car model. A Proportional-Integral-Derivative (PID) controller was designed to actively control suspension forces, achieving significant improvements in ride comfort and handling characteristics. Comprehensive frequency and time domain analysis demonstrates that the active suspension system achieves a peak magnitude reduction of 15.24 dB compared to the passive system, translating to an 82% reduction in vibration amplitude. Time domain simulations show 50% reduction in peak body acceleration and 62.5% faster settling time, validating the effectiveness of active control strategies in enhancing vehicle suspension performance.

Keywords: Active suspension, PID control, Quarter-car model, MATLAB/Simulink, Frequency response analysis

1 Introduction and Objectives

1.1 Project Overview

Vehicle suspension systems are critical for passenger comfort, vehicle stability, and road handling. While traditional passive suspension systems use fixed spring-damper characteristics, active suspension systems incorporate actuators and control systems that can adapt to varying road conditions in real-time. This project develops and evaluates an active suspension system using PID control for a quarter-car model.

1.2 Key Objectives

1. Develop a mathematical model of a quarter-car suspension system using transfer functions and state-space representations
2. Design a PID controller for active suspension control with velocity feedback
3. Implement the system in MATLAB/Simulink environment
4. Perform comprehensive frequency and time domain analysis
5. Compare performance metrics between passive and active suspension configurations

2 System Model and Parameters

2.1 Quarter-Car Model

The quarter-car model represents the vertical dynamics of one corner of a vehicle, consisting of:

- **Sprung mass (m_s):** Vehicle body (300 kg)
- **Unsprung mass (m_{us}):** Wheel assembly (40 kg)
- **Suspension spring/damper:** $k_s = 18\,000$ N/m, $c_s = 1200$ N s/m

- **Tire spring/damper:** $k_t = 195\,000\text{ N/m}$, $c_t = 75\text{ N s/m}$
- **Active control force:** F_c (variable, limited to $\hat{A}\pm 5000\text{ N}$)

2.2 Mathematical Representation

The system dynamics are governed by Newton's second law applied to both masses:

Sprung mass equation:

$$m_s \ddot{x}_s = -k_s(x_s - x_{us}) - c_s(\dot{x}_s - \dot{x}_{us}) + F_c \quad (1)$$

Unsprung mass equation:

$$m_{us} \ddot{x}_{us} = k_s(x_s - x_{us}) + c_s(\dot{x}_s - \dot{x}_{us}) - k_t(x_{us} - x_r) - c_t(\dot{x}_{us} - \dot{x}_r) - F_c \quad (2)$$

where x_s , x_{us} , and x_r represent sprung mass, unsprung mass, and road profile displacements respectively.

3 Controller Design

3.1 PID Control Strategy

The active suspension employs a PID controller with velocity feedback. The control force is generated based on the error between desired (zero) and actual body velocity:

$$F_c(s) = C_{PID}(s) \cdot (0 - s \cdot X_s(s)) \quad (3)$$

The PID controller structure with derivative filter is:

$$C_{PID}(s) = K_p + \frac{K_i}{s} + \frac{K_d N s}{s + N} \quad (4)$$

3.2 Controller Gains

Through iterative tuning to optimize performance while maintaining stability, the following gains were selected:

Parameter	Value
Proportional gain, K_p	5,000
Integral gain, K_i	100
Derivative gain, K_d	2,000
Filter coefficient, N	100

4 Implementation and Methodology

The system was implemented in MATLAB/Simulink with modular subsystems including:

- Road input generation (step, sinusoidal, random)

- Tire and suspension dynamics blocks
- Mass dynamics with double integration
- PID controller with saturation and rate limiting
- Data acquisition and visualization

Analysis methods included:

- Bode plot analysis for frequency response characterization
- Time-domain simulations for step and sinusoidal road inputs
- Performance metrics calculation (acceleration, settling time, suspension travel)

5 Results and Performance Analysis

5.1 Frequency Domain Results

Bode plot analysis revealed significant improvements in the active system:

Table 1: Frequency response comparison

Metric	Passive	Active
Peak magnitude (dB)	5.98	−9.26
Peak frequency (Hz)	1.21	1.34
Magnitude improvement (dB)	–	15.24
Amplitude reduction (%)	–	82.0

The 15.24 dB peak reduction demonstrates superior vibration isolation across the critical frequency range (0.5-10 Hz) covering typical road disturbances.

5.2 Time Domain Results

Step input simulations demonstrated substantial improvements:

Table 2: Time domain performance metrics

Metric	Passive	Active
Peak body acceleration (m/s^2)	12.0	6.0
Acceleration reduction (%)	–	50.0
Settling time (s)	4.0	1.5
Settling time reduction (%)	–	62.5
Maximum suspension travel (m)	0.08	0.05
Travel reduction (%)	–	37.5

5.3 Key Observations

- **Vibration Suppression:** 82% reduction in vibration amplitude significantly improves passenger comfort
- **Transient Response:** 62.5% faster settling eliminates sustained oscillations
- **Body Acceleration:** 50% reduction directly enhances ride quality
- **Suspension Utilization:** 37.5% reduction in travel improves stroke efficiency
- **Stability:** System remains stable with adequate gain and phase margins

6 Conclusions and Future Work

6.1 Key Achievements

This project successfully demonstrated:

1. Effective mathematical modeling of quarter-car suspension dynamics
2. PID controller design achieving substantial performance improvements
3. Comprehensive validation through frequency and time domain analysis
4. Practical implementation in MATLAB/Simulink environment

The quantitative results validate that active suspension with PID control provides:

- Superior vibration isolation (82% amplitude reduction)
- Enhanced passenger comfort (50% lower peak acceleration)
- Faster disturbance rejection (62.5% settling time reduction)
- Better suspension utilization (37.5% travel reduction)

6.2 Practical Implications

The achieved improvements would translate to noticeably better ride comfort and handling in real-world driving conditions. The modular Simulink implementation provides an excellent platform for exploring advanced control strategies and system enhancements.

6.3 Future Extensions

Potential areas for further development include:

- Advanced control strategies (LQR, MPC, adaptive control, H-infinity)
- Full vehicle (7-DOF) model with pitch and roll dynamics
- Hardware-in-the-loop testing and experimental validation
- Semi-active solutions using magnetorheological dampers
- Preview control using forward-looking sensors

6.4 Learning Outcomes

This project provided hands-on experience with:

- Mathematical modeling of complex mechanical systems
- Controller design, tuning, and performance evaluation
- MATLAB/Simulink simulation and analysis
- Frequency and time domain analysis techniques
- Technical documentation and scientific report writing

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